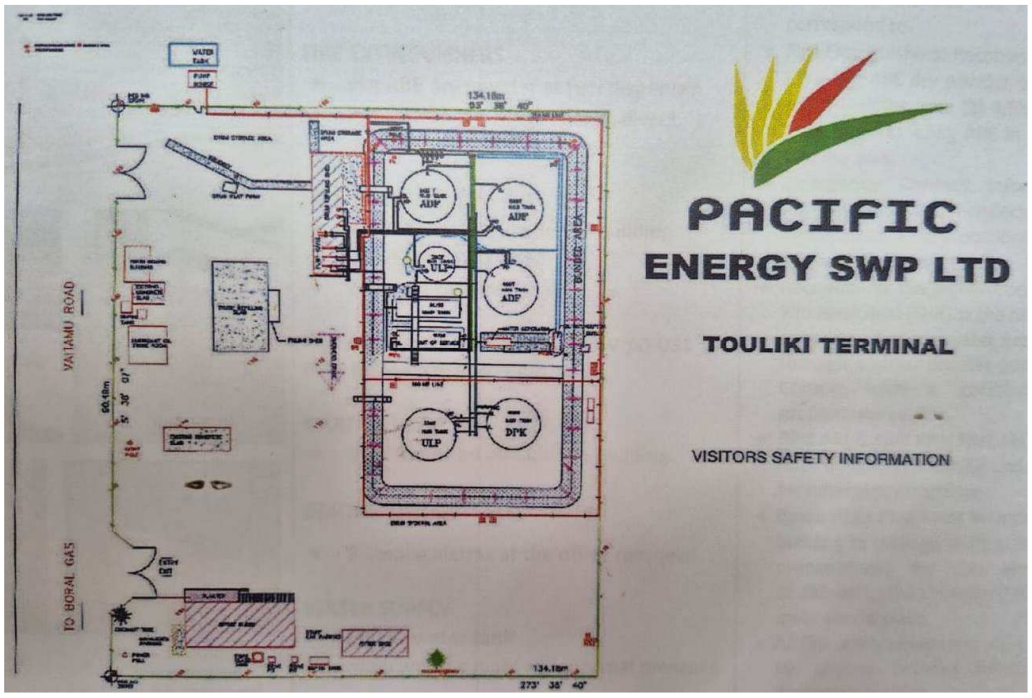
SEA-2023-24656: TONGA
Technical Assistance - energy sector

PETROLEUM STORAGE AUDITING REPORT



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02				
03				

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SEA-2023-24656: TONGA
Technical assistance to the energy sector

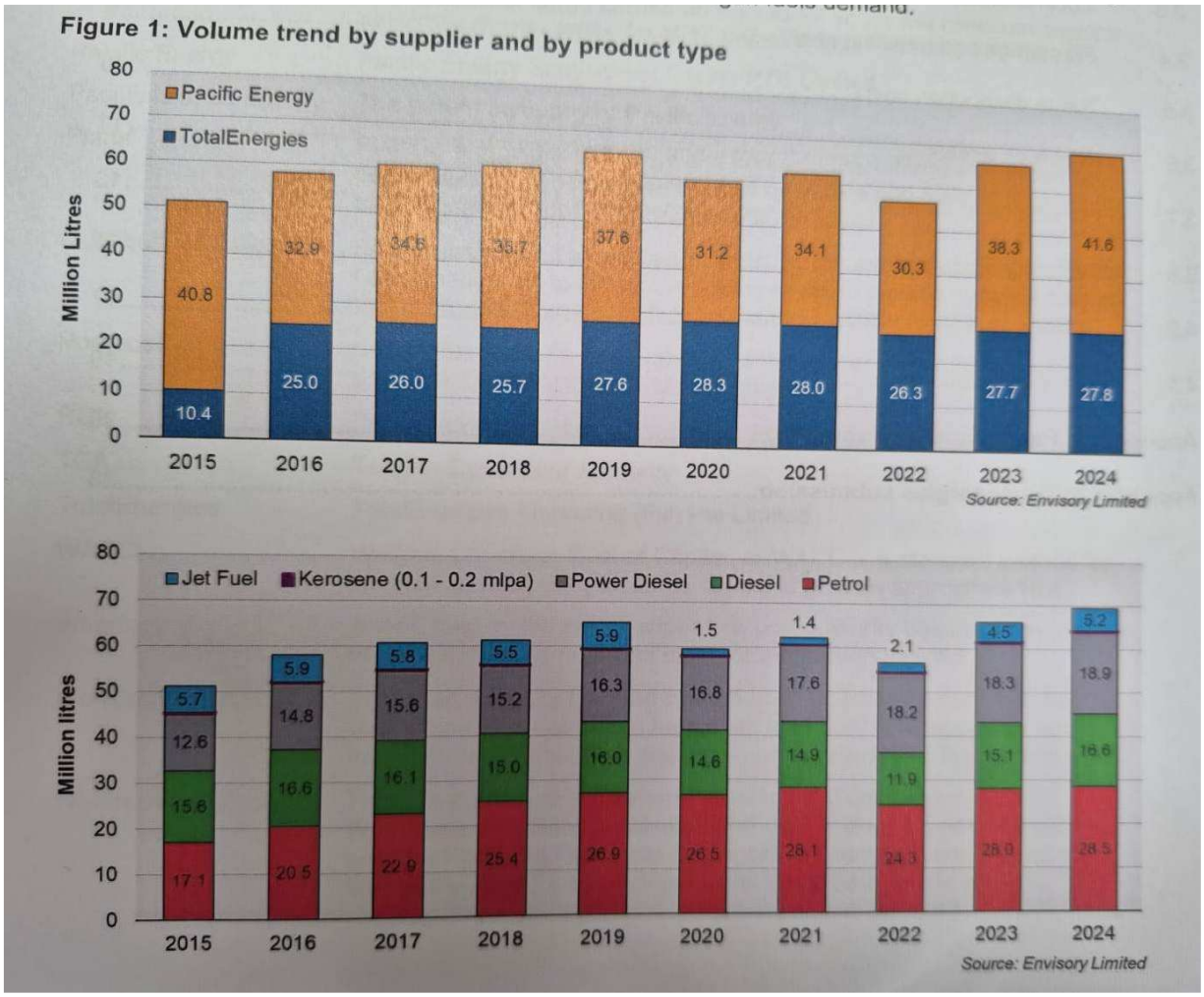
Petroleum storage Auditing Report

1. *Introduction:*

On the 10th of October 2025 the Task force" two staff from each MoTED, MoI, Health and Fire Brigade and DoE inspected the Pacific Energy and Total Energies' main storage.

The purpose of the EU-Tech Assistance Team Leader involvement was to assist the MoTED inspection team to observe the safety and operational practices of both suppliers and understand the recent supply challenges based on experience.

The following table of 2015- 2024 distribution total volume and, by product for both companies, Pacific Energy and Total Energies. Shows the percentage of supply to Petroleum products to all Tonga islands. These findings are part of the Competent Authority annual report.



2. Findings:

2.1The Pacific Energy in Tonga

The Tonga operations are part of the independent service providers group (Fiji, American Samoa, Tonga, Tuvalu, Cook, Kiribati and Vanuatu, Papua New Guinea) registered in French Polynesia. In Tonga they operate three supply depots at the airport, 19 service stations with 22 employees and storage of about 41.6 million litres per year.

Pacific Energy supplies about 80% of the energy market, providing Jet fuel, kerosene, ULP and diesel fuel for all islands in Tonga. They have three main storage facilities and one at the Tongatapu airport for Jet (A1) Fuel. The company has about 22 staff managing the storage facilities and transport to services stations. In 2024 supply about 19 service stations on several islands of Tonga supplying about 41.6 M litres of ULP, Jet fuel and small amount of Diesel.

The Pacific Energy Group espouses a commitment of quality service, safety and minimizing environmental risks. The inspection was conducted with the support a Pacific Energy Plant Manager. The purpose was to present the policies addressing the company's commitments to the Task Force. The policies were posted on the staff-office walls but were hanging high so hard to read. Signage inside and on the fence indicating specific safety requirements for visitors to enter the premises. However, the MoTED team noticed maintenance and fuel trucks entering and exiting without any inspections or monitoring. Both main gates were continuously open during our presence. All restricted signage were very bleached and old; the red warnings were no longer obvious and hard to read. The Safety Policy stated that full personal protective equipment (PPE) is to be used while in the compound. Yet the helmets were not rated as safety helmets and staff did not wear gloves.

The firefighting included foam for the tank storage and powder fire extinguishers, all certified by a contracted company and to date. The foam storage was 400 litres positioned in three locations around the tanks. The foam reserve was limited and mentioned that more new drums are due to arrive. The fire hydrant is connected to a saltwater storage and pumping station in the Total Energy yards. When the alarm goes on the Total Energy staff will manually start up the pumping station for the Pacific Energy staff. The hydrant is flush during training exercises every month but has only one outlet. The condition is very poor, and no record of maintenance was presented. It was obvious that some of the repainted valves had paint connecting each moving part. The tank containment areas had water from recent rain but was not drained which may reduce the volume of containment. There was no indication that the fire hydrant was connected to the top of the tanks.

The gantry is very small but enough for one operator. The condition of the equipment is again showing poor maintenance and replacement plan. The earthing system is present but not tested. There was no mention of emergency shut off and the ability to stop the operation if the earthing is not conducting correctly. The operator (driver) at the time of the inspection was not wearing safety harness and questioned the steps taken to ensure safety during the filling of the tanker. Once finished the truck blocked the exit gate to carry out the paperwork in the office.

Most of the facility shows signs of lack of maintenance and dilapidated and neglected equipment. The handrails above the tank are so rusted that some of them have been replaced with ropes. The stairs are rusted looking from a distance and would need to be checked for strength. The valves are operational but again it is obvious that paint is hiding a lot of the actual conditions.

The spill kits are very limited and the emergency trailer for spill response was empty. The reason given was a new one is on its way. When asked about spillage response the comment made was that the containment areas working, and the sumps are drained with a mobile pump.

Two sludge tanks in the containment areas are out of commission and will be removed shortly. No sign of new ones to be installed. There is space for another tank to extending the storage capability. The comment was made that facility had plenty of land and could expend the storage. A proposal was submitted but no news on the new development. At the end of the visit, we requested copies of the operational check lists.

In overall the Pacific Energy is operating with a limited maintenance program and the facility is showing sign of risks for failure. The policies are in place and regular training is in place as we were told. However, the short time during the visit shown that standards practices are not in place.

2.2 Total Energies

Specifically, the Total Energies storage facility in Tongatapu. At the time of this report the data for the company storage capacity was not shared. The company is planning further investment to improve and increase its storage capacity. Four 7Kl tanks will be decommissioned and transferred to TPL. Seven 12Kl new underground tanks will be added to the total capacity storage. At this stage there is no confirmation of what type of product these new tanks will store.

The Total Energies operations are part of Fiji consortium and overseen by management in Fiji. The facility is well maintained and all conditions for quality service, safety and minimizing environmental risks are well advertised and all staff are complying.

The facility has some Supervisory Control and Data Acquisition system (SCADA) to ensure situations like overfilling of tanks with emergency shut of valves. The facility is also equipped with sensors within the containment areas to monitor fuel spills and excess vapours, which are connected to alarms. The gantry has an earthing, and valves fill-control. If the earthing is not connected properly the fuelling will not start. Safety harnesses are also available.

The storage facility is well maintained, and operational maintenance practices are in place. For example, all valves are numbered, and a maintenance programme is in place. All. The tank bunkering was dry during the visit of both facilities. All painting is recent, and all signs are clearly visible and identified. We witness at least four spill kit stations, a fully equipped trailer for emergency spills and a trailer for spill from ship leaks during bunkering. The fire system consists of three high volume pumps dispensing from a 10Kl seawater tank that is also supporting the Pacific Energy facility for fire emergencies.

Visual observations indicate that Total Energies is better equipped and managed to minimize such risks compared to Pacific Energy. It is also evident that Total Energies continues to invest in enhancing storage safety and capacity, whereas Pacific Energy has shown less diligence in that area. It is important to note, as highlighted in the Competent Authority's pricing justification, that Pacific Energy holds a larger share of supply and achieves a higher return on investment, whereas Total Energies continuous investment has a lower return on investment.

Based on the June 2024 Annual Review of Fuel Pricing for Tonga the Competent Authority assessment with the support of Envisory Consultants. The figures below support the inspection findings.

They're increasing evidence that without additional investment on operational and increase in storage, the challenges to maintaining a constant chain of supply will worsen for Pacific Energy. The fact that there are only six days of ULP stock could be resolved easily with additional storage for both companies.

There are concerns in Envisory Consultants reports about the losses reported by Pacific Energy and the lack of identification of where the losses are occurring. The industry norm is 24% where Pacific Energy reported 30%.

3. Recommendations

Considering the brief visits to both facilities, a more detailed assessment of each company's operational practices, is required to determine their capability to contain any potential catastrophic event.

A final report is required to define the current status of fuel storage capacities and operational practices. Based on the findings, the Ministry of Trade and Economic Development (MoTED) and the Ministry of Energy (MEIDECC) should develop a Memorandum of Agreement (MoA) outlining

their respective responsibilities and areas of collaboration to address the necessary reforms aimed at reducing safety risks and mitigating storage shortages. Furthermore, both Ministries should agree on a consultation process with Stakeholders to review the final report and implement the recommended reforms.

The Competent Authority currently holds contracts with both companies. It is recommended that these contracts be revised to ensure that the necessary operational improvements are effectively addressed. While the existing contracts may be primarily focus on business registration and pricing conditions, they do not adequately cover operational practices. The revised contracts should therefore include minimum requirements for annual investment and a clear plan for increasing storage capacity in line with demand, ensuring a reliable and continuous fuel supply. Furthermore, the contracts should incorporate a licensing process aligned with the *Energy Service Provider Licensing Requirements* to ensure that safety standards and operational practices are properly regulated and monitored in the delivery of services to customers.

3.1Next Steps

To get a better understanding of the situation, it is recommended that the following steps are taken to provide a detailed assessment to report to the Government of Tonga.

- Develop a ministerial MoA between MoTED and MEIDECC.
- Consultations with Stakeholders (Petroleum Companies, customers, outer islands Authorities etc..).
- Inspections of all storage terminals for Pacific Energy and Total Energies on all islands.
- Identify the actual usable storage capacity of each products stored
- Identify daily supply of all products at each location.
- Identify the shipping schedules, duration of the deliveries and quantities discharged at each location.
- Identify the current storage conditions and outline the plan for replacement or expansion of storage facilities.
- Identify the equipment conditions, and risks of non-compliance.
- Identify operational practices and compliances.
- Review current contractual terms and conditions between corporate arrangements, between both companies and government.

3.2Final report

- Recommendations on improvement of each storage facilities for supply expectations and reduce risk of outages.
- Recommendations changes to practices on receipts and deliveries of each product to ensure adequate reserves within the shipment schedules.
- Recommendations on the companies structuring an asset replacement and upgrade plan to ensure ongoing safety, compliance, and operational efficiency.
- Recommendations on contractual agreements and implementation of regulatory frameworks to ensure compliance contractual agreements.

4. *Annex - 1*

2.1 Supplier cost and profitability trend

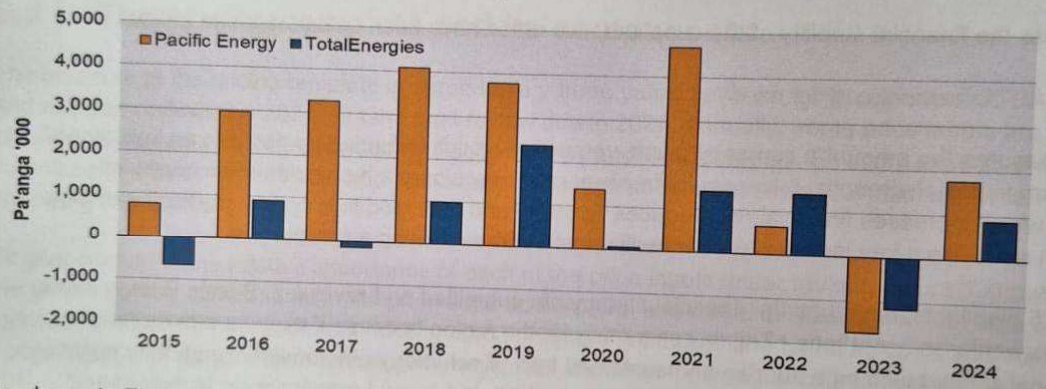
To provide context to the discussions in Section 3.0 on the recommended changes to the pricing template inputs, we have reviewed the financial situation of Pacific Energy and TotalEnergies.

Compared to 2023, where both companies made a loss (Figure 2), both have returned to making a net profit (after tax) in 2024, although lower than was the case pre-COVID:

- Pacific Energy NPAT: TOP 1.760 million
- TotalEnergies NPAT: TOP 0.889 million

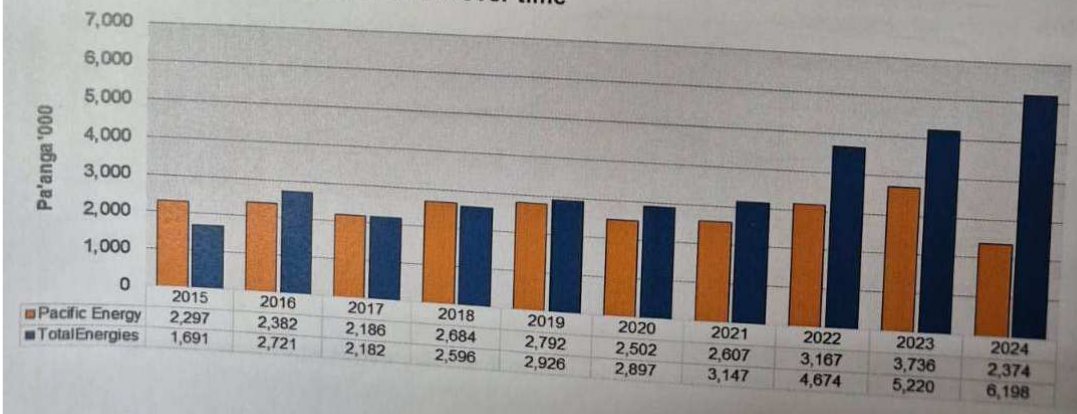
In 2024, fuels prices (for all fuels except power diesel) were set by Pacific Energy's price, as the lowest cost supplier, although some items in Pacific Energy's price submission were not approved by TCA, which will have reduced their profit, compared to what Pacific Energy was seeking.

Figure 2: Comparison of supplier profitability over time



As shown in Figure 3, Pacific Energy has reduced its OPEX by TOP 1.362 million driven by a reduction in "other" operating expenses, this category of expense was relatively high over the last couple of years with 2024 now back to similar levels seen in 2022-2023. TotalEnergies OPEX increased by TOP 0.978 million due to an increase in selling, marketing and distribution expenses.

Figure 3: Comparison of supplier OPEX over time



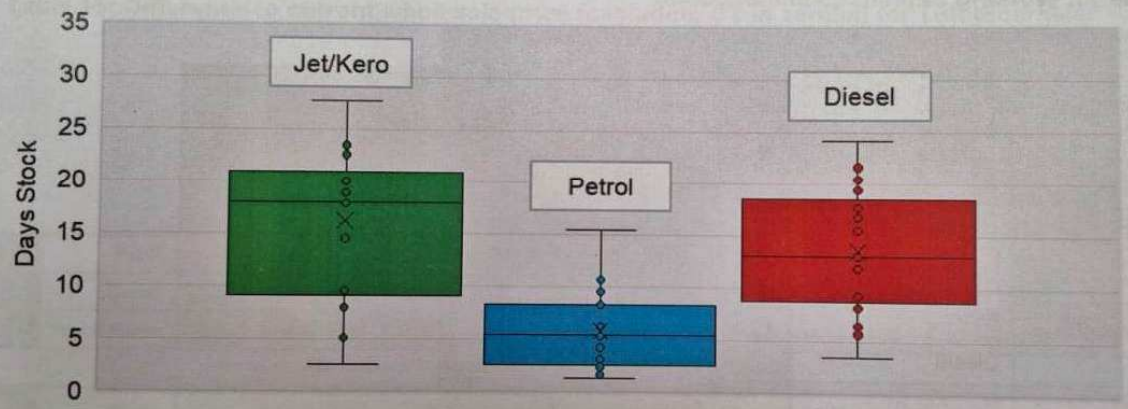
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Pacific Energy	2,297	2,382	2,186	2,684	2,792	2,502	2,607	3,167	3,736	2,374
TotalEnergies	1,691	2,721	2,182	2,596	2,926	2,897	3,147	4,674	5,220	6,198

Pacific Energy' Fiji stock holding for unleaded petrol increased from 49 days to 60 days, while TotalEnergies Fiji diesel stockholding increased from 43 to 54 days. These two items should be held constant at the 2024 approved levels unless further information is provided.

Tonga supply resilience based on the stock days

While not part of the annual review, Envisory has assessed Tonga's resilience to supply disruption for Pacific Energy based on their LCT cargo data supplied for their stockholding submission. Similar analysis was not possible for TotalEnergies due to the lack of submitted evidence, but their supply situation is similar to Pacific Energy, so this analysis is representative of the situation for Tonga.

Figure 10: Pacific Energy's remaining days stock on vessel arrival for Tonga



The key findings were:

- On average there is less than six days petrol stock in Tonga at replenishment, giving little room for delays, for events such as late cargoes or poor weather.
- Whereas diesel has (on average) around 13.5 days stock and jet/kerosene has around 16 days stock, meaning there is more resilience to disruption than observed for petrol.

The TCA may want to investigate if there should be some minimum stockholding obligation for suppliers to improve supply resilience for Tonga.

Losses

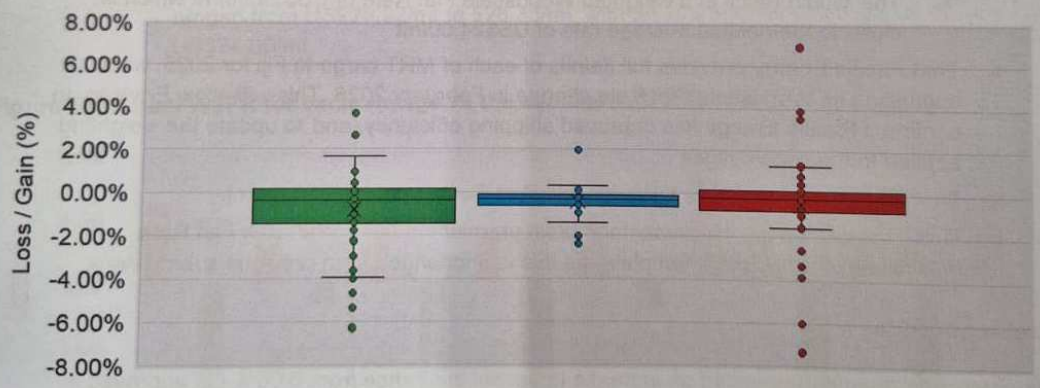
Pacific Energy has provided loss information for its 2024 deliveries, showing overall losses of 0.24% for MRT voyages to Fiji, and 0.40% for LCT voyages to Tonga. The approved losses for both MRT and LCT in 2024 were 0.30%.

- Pacific Energy's performance on MRT losses has been positive with these reducing from 0.75% in 2022, 0.42% in 2023 and now 0.24% in 2024. While losses are now below the TCA approved level of 0.30% it is still recommended to maintain the losses in the fuel price template at this rate as this is considered to be industry best practice.
- Despite improvements in LCT losses (2022: 0.75% → 2023: 0.42% → 2024: 0.40%) they are still above industry best practice. An analysis of the data provided shows there are some significant outliers.

LCT Losses

Figure 12 shows the reported 2024 loss/gains on diesel and to a lesser extent petrol have some very significant outliers. Diesel discharges have been in the range -/+8% which is unacceptably high, and petrol discharge losses have also ranged from -6% to 4% which is also very concerning. Whereas the losses and gains on kerosene/jet have been relatively tight.

Figure 12: Loss / gains by product type



As shown in Table 16, drilling into the data further highlights a problem at Vava'u where there are significantly higher product losses than other ports. Losses for kerosene/jet and diesel at Savusavu are also of concern however they only form a small proportion of overall outturn so have less impact on the total loss/gains. Excluding Vava'u from the loss/gain calculations for 2024 deliveries results in an overall loss of 0.22% which is consistent with both the MRT losses, and industry best practice.

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[14] Change calculated up to Q1 2024 to reflect costs borne during 2024. This was at the peak of the insurance costs increases, from that point, costs started decreasing.

[15] Marsh Global Insurance Market Index, Pacific Insurance rates.